Physics

Kinematics

Grade 7

Quarter 1

Name: _____ Class: _____ Date: _____

Designed by Sunquist in Park Lane Harbor

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Unit 2 — Kinematics

Kinematics — Position and Reference Frames

Lesson 1 of 5

Describe position with 1-D and 2-D coordinate systems and frames of reference, add vectors, and distinguish distance from displacement.

What is Kinematics?

- **Kinematics** is the study of **motion** — how things move.
- We describe motion using **words, numbers, equations, and graphs**.
- We will study **1-D motion** (along a straight line); we also use **2-D coordinates** to describe position and direction.

Kinematics Variables and Symbols

Quantity	Symbol	Units	Type
position	x	m	vector →
distance	d	m	scalar
displacement	Δx	m	vector →
time	t	s	scalar
speed	s	$\frac{\mathrm{m}}{\mathrm{s}}$	scalar
velocity	v	$\frac{\mathrm{m}}{\mathrm{s}}$	vector →
acceleration	a	$\frac{\mathrm{m}}{\mathrm{s}^2}$	vector →

Kinematic Equations

- $\bar{v} = \frac{\Delta x}{t}$
- $s = \frac{d}{t}$
- $a = \frac{\Delta v}{t} = \frac{v - v_0}{t}$
- $\Delta x = x - x_0$
- $v = v_0 + at$
- $\Delta x = v_0 t + \frac{1}{2}at^2$
- $v^2 = v_0^2 + 2a\Delta x$

Scalars and Vectors

> **Scalar**
> A quantity with **magnitude** (a number and unit) only.

$$t,\ d,\ s$$

> **Vector**
> A quantity with **magnitude** and **direction**.

$\vec{x}$ $\vec{\Delta x}$ $\vec{v}$ $\vec{a}$

Vector Addition

> **Tip**
> Vectors must be on the same axis (collinear): **same** direction **add**, **opposite** directions **subtract**.

Same Direction $\overrightarrow{5\ \mathrm{m}} + \overrightarrow{3\ \mathrm{m}} = \overrightarrow{8\ \mathrm{m}}$

Opposite Direction $\overrightarrow{5\ \mathrm{m}} + \overleftarrow{3\ \mathrm{m}} = \overrightarrow{2\ \mathrm{m}}$

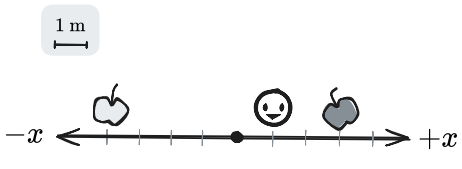
Position

> **Position**
> The location of an object in a **coordinate system** or relative to a **reference point**. Position, x, is a vector and its unit is m

A **coordinate system** has:

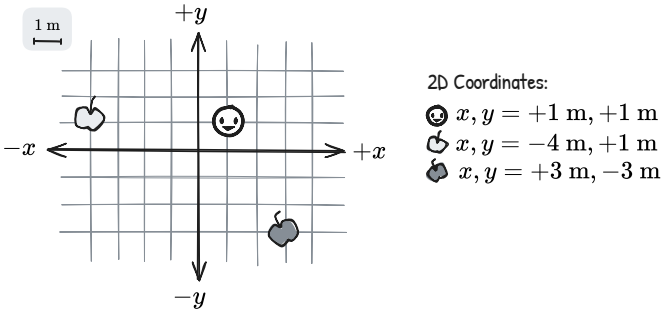
- **axes** (show positive and negative directions)
- **origin** (the center of the coordinate system)
- **scale** and **units**

1-D Coordinate System



Using the face as a **reference point**, there is one fruit 5 meters left of the face and another fruit 2 meters right of the face.

2-D Coordinate System

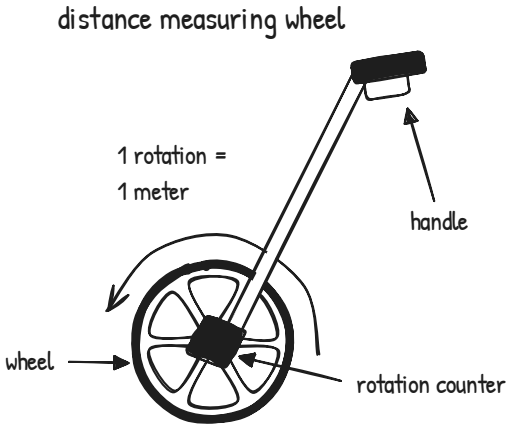


There is one fruit 5 meters left of the face and another fruit 2 meters right and 4 meters below the face.

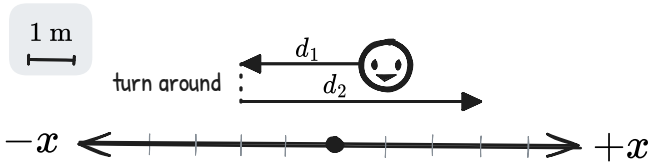
Distance

> **Distance**
> The length of the path taken. Distance, d, is a **scalar** and its unit is m.

Distance is always positive. An object's distance increases every time it changes position, regardless of direction.



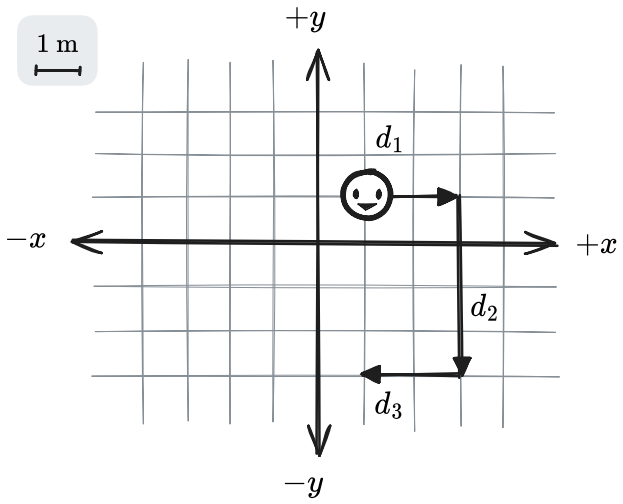
Distance in 1-D



$$d_{total} = d_1 + d_2$$

$$d_{total} = 3\,\mathrm{m} + 5\,\mathrm{m} = 8\,\mathrm{m}$$

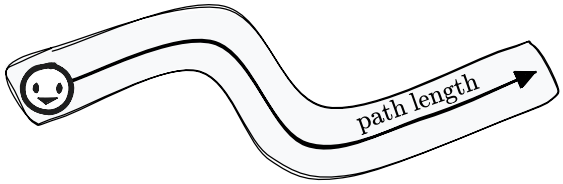
Distance in 2-D



$$d_{total} = d_1 + d_2 + d_3$$

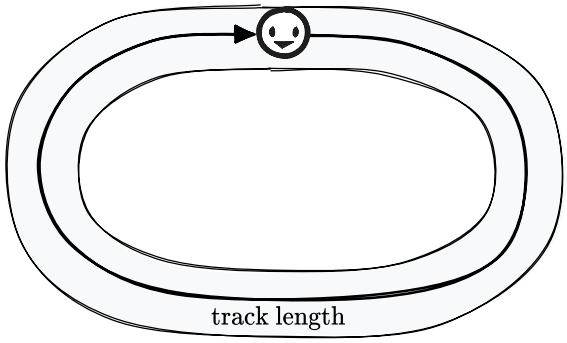
$$d_{total} = 2\,\mathrm{m} + 4\,\mathrm{m} + 2\,\mathrm{m} = 8\,\mathrm{m}$$

Path Distance



$$d = \text{path length}$$

Track Distance



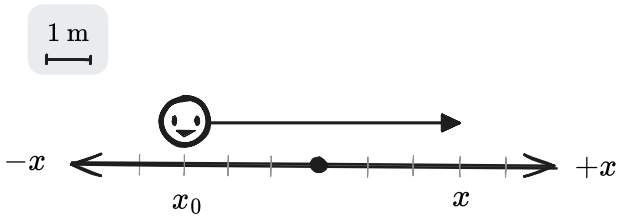
$$d = \text{number of laps} \times \text{track length}$$

Displacement

> **Displacement**
> A change in position. Displacement, Δx, is a **vector** and its unit is m.

- $\Delta x = x - x_0$, where x is the *final position* and x_0 is the *initial position*.
- $\Delta x_{total} = \Delta x_1 + \Delta x_2 + \cdots + \Delta x_n$

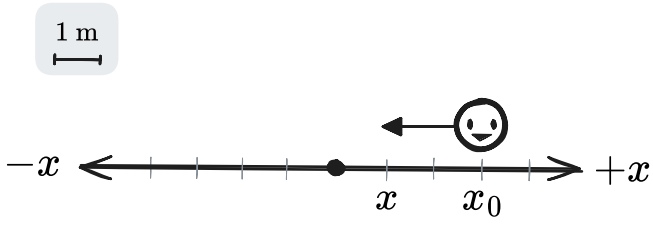
1-D Displacement



$$\Delta x = x - x_0$$

$$\Delta x = 3\,\mathrm{m} - (-3\,\mathrm{m}) = 6\,\mathrm{m\,right}$$

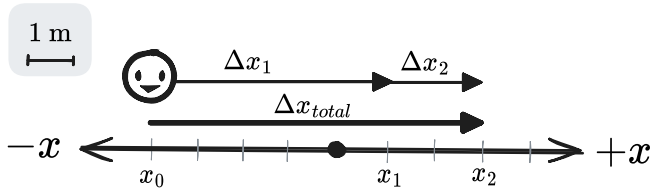
1-D Displacement (negative)



$$\Delta x = x - x_0$$

$$\Delta x = 1\,\mathrm{m} - 3\,\mathrm{m} = -2\,\mathrm{m} = 2\,\mathrm{m\,left}$$

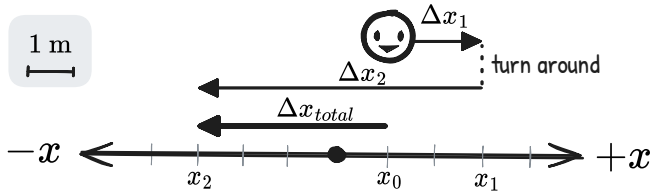
Total 1-D Displacement



$$\Delta x_{total} = \Delta x_1 + \Delta x_2 = 5\,\text{m} + 2\,\text{m} = 7\,\text{m right}$$

$$\Delta x_{total} = x_2 - x_0 = 3\,\text{m} - (-4\,\text{m}) = 7\,\text{m right}$$

Resultant 1-D Displacement (+/-)



$$\Delta x = \Delta x_1 + \Delta x_2 = 2\,\text{m} - 6\,\text{m} = -4\,\text{m} = 4\,\text{m left}$$

$$\Delta x = x_2 - x_0 = -3\,\text{m} - 1\,\text{m} = -4\text{m} = 4\,\text{m left}$$

Distance vs Displacement

Distance	Displacement
total path length	change in position
scalar (no direction)	vector (magnitude + direction)
never negative	can be positive, negative, or zero
depends on the route taken	depends only on start and finish

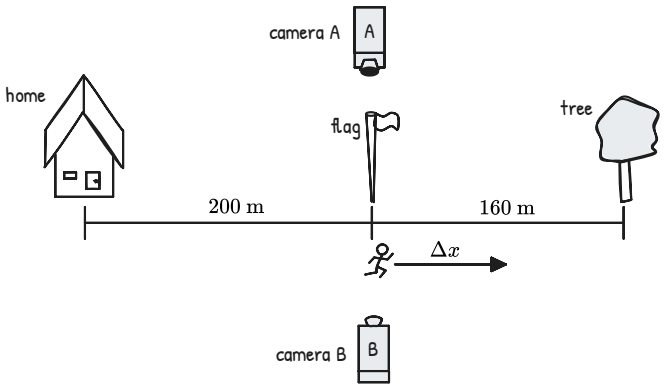
Frames of Reference

> Frame of Reference
> The view from which motion is measured.

Every measurement is made **relative to** a **reference point** — so the **same motion can look different** from different frames of reference.

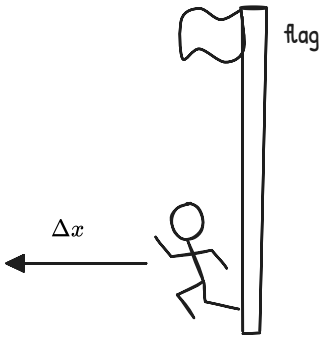
Frame of Reference Example

The flag is the **reference point**. Camera A and Camera B record the man running in opposite directions from the flag.



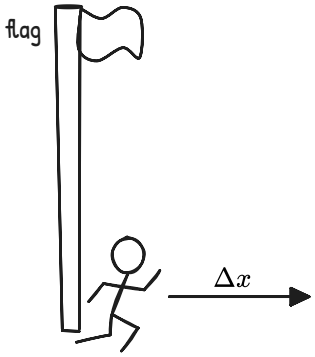
Camera A's View

In this view, the man is running to the left of the flag.



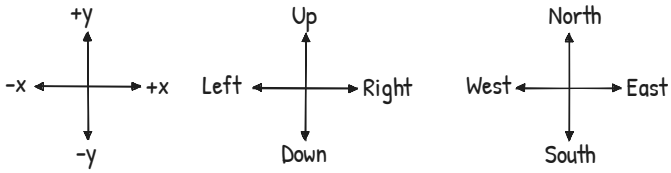
Camera B's View

In this view, the man is running to the right of the flag.



"Positive" and "Negative" Directions

The four directions correspond to positive and negative as shown:



Kinematics — Speed and Velocity

Lesson 2 of 5

Calculate speed and velocity, and combine velocity vectors to find resultant and relative velocities.

Time

> Time
> The duration between two instants. An instant is a single point on a timeline. Time, t, is a scalar measured in s

- *A sprinter ran from the starting blocks to the finish line in* $10.5\,\text{s}$.
- *A car braked from* $25\ \frac{\text{m}}{\text{s}}$ *to* $0\ \frac{\text{m}}{\text{s}}$ *over* $4\,\text{s}$.

Speed

> **Speed**
> How fast something moves. Speed, s, is a scalar measured in $\frac{\text{m}}{\text{s}}$.

- $s = \frac{d}{t}$.
- *The speed of the bus is* $40 \frac{\text{m}}{\text{s}}$.

Velocity

> **Velocity**
> How fast something moves *and in what direction*. Velocity, v, is a vector $\frac{\text{m}}{\text{s}}$.

- $v = \frac{\Delta x}{t}$.
- *The velocity of the bus is* $40 \frac{\text{m}}{\text{s}}$ east.

Problem Solving Steps

1. **Annotate the problem to find the quantities.**
2. **What equation connects the quantities?**
3. **Plug in with units and solve.**
4. **Check the units and direction (if applicable).**

Example: Speed

A train travels 2000 m in 60 s. What is its speed?

Solution

A train travels 2000 m in 60 s. What is its speed?

$$s = \frac{d}{t} = \frac{2000 \text{ m}}{60 \text{ s}} = 33.3 \frac{\text{m}}{\text{s}}$$

s: $\frac{m}{s}$, scalar

Example: Distance

A dog runs at a constant speed of $5 \frac{m}{s}$ for a time of 30 s. What is the distance traveled?

Solution

A dog runs at a constant speed of $5 \frac{m}{s}$ for a time of 30 s. What is the distance traveled?

$$d = st = 5 \frac{m}{s} \cdot 30 \text{ s} = 150 \text{ m}$$

d: m, scalar

Example: Velocity

A bird flies 500 m east in 25 s. What is its velocity?

Solution

A bird flies 500 m east in 25 s. What is its velocity?

$$v = \frac{\Delta x}{t} = \frac{500 \text{ m}}{25 \text{ s}} = 20 \frac{\text{m}}{\text{s}} \text{ east}$$

v: $\frac{\text{m}}{\text{s}}$, east

Example: Displacement

A balloon floats upward at a constant velocity of $2.5 \frac{\text{m}}{\text{s}}$ for a time of 20 s. What is its displacement?

Solution

A balloon floats upward at a constant velocity of $2.5 \frac{\text{m}}{\text{s}}$ for a time of 20 s. What is its displacement?

$$\Delta x = vt = 2.5 \frac{\text{m}}{\text{s}} \cdot 20 \text{ s} = 50 \text{ m up}$$

Δx: m, up

Resultant Velocity

> **Resultant Velocity**
> The sum of two or more velocity vectors acting on the **same object**.

$$v_{resultant} = v_A + v_B$$

Resultant Velocity (Same Direction)

A boat moves $3 \frac{\text{m}}{\text{s}}$ east while the current pushes $1 \frac{\text{m}}{\text{s}}$ east.

$$v_{\text{resultant}} = 3 \frac{\text{m}}{\text{s}} + 1 \frac{\text{m}}{\text{s}} = 4 \frac{\text{m}}{\text{s}} \text{ east}$$

Resultant Velocity (Opposite Direction)

A boat moves $3 \frac{\text{m}}{\text{s}}$ east while the current pushes $1 \frac{\text{m}}{\text{s}}$ west.

$$v_{\text{resultant}} = 3 \frac{\text{m}}{\text{s}} - 1 \frac{\text{m}}{\text{s}} = 2 \frac{\text{m}}{\text{s}} \text{ east}$$

Relative Velocity

> **Relative Velocity**
> The velocity of one object measured from the **frame of reference** of another object

Velocity of object B relative to object A: $v_{BA} = v_B - v_A$

Velocity of object A relative to object B: $v_{AB} = v_A - v_B$

Relative Velocity (Same Direction)

Two ships move east: ship A at 10 m/s and ship B at 15 m/s

Velocity of B relative to A:

- $v_{BA} = v_B - v_A = 15 \frac{\text{m}}{\text{s}} - 10 \frac{\text{m}}{\text{s}} = 5 \frac{\text{m}}{\text{s}}$ east.

Velocity of A relative to B:

- $v_{AB} = v_A - v_B = 10 \frac{\text{m}}{\text{s}} - 15 \frac{\text{m}}{\text{s}} = 5 \frac{\text{m}}{\text{s}}$ west.

Relative Velocity (Opposite Directions)

Ship A at $10\ \frac{\text{m}}{\text{s}}$ east and ship B at $15\ \frac{\text{m}}{\text{s}}$ west.

Assign east as positive and west as negative:

- $v_A = +10\ \frac{\text{m}}{\text{s}}, \quad v_B = -15\ \frac{\text{m}}{\text{s}}$

Velocity of B relative to A:

- $v_{BA} = v_B - v_A = -15\frac{\text{m}}{\text{s}} - 10\frac{\text{m}}{\text{s}} = -25\frac{\text{m}}{\text{s}} = 25\frac{\text{m}}{\text{s}}$ west.

Velocity of A relative to B:

- $v_{AB} = v_A - v_B = 10\ \frac{\text{m}}{\text{s}} - (-15)\ \frac{\text{m}}{\text{s}} = +25\frac{\text{m}}{\text{s}} = 25\frac{\text{m}}{\text{s}}$ east.

Kinematics — Speed, Velocity, and Acceleration

Lesson 3 of 5

Distinguish constant, average, and instantaneous speed and velocity, and describe acceleration as speeding up, slowing down, or changing direction.

Average Speed

> **Average Speed**
> Total distance divided by total time. This is useful when there is more than one motion interval with distances and times.

$$s_{\text{average}} = \frac{d_{\text{total}}}{t_{\text{total}}}$$

Average Velocity

> **Average Velocity**
> Total displacement divided by total time. Total displacement is calculated using **vector addition**.

$$v_{\text{average}} = \frac{\Delta x_{\text{total}}}{t_{\text{total}}}$$

Constant Speed and Constant Velocity

> **Constant Speed and Constant Velocity**
> Speed is constant when the rate of distance traveled never changes. Velocity is constant when both its magnitude and direction never change.

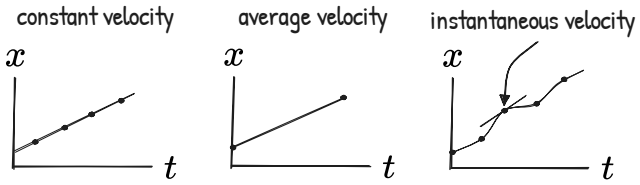
- Constant at every instant t.
- A straight line (constant slope) on a position-time graph.
- Velocity is constant only when moving in a straight line at constant speed.

Instantaneous Speed and Velocity

> **Instantaneous Speed and Velocity**
> Speed at a single instant (a point on a timeline), such as initial, final, or at a specific time. Velocity is the same, but with direction.

- Instantaneous speed corresponds to the slope of the position-time graph at that instant.

Constant vs Average vs Instantaneous



- slope = velocity
- constant velocity has a constant slope
- average velocity is the slope between the final and initial positions
- instantaneous velocity may be different at every time

Acceleration

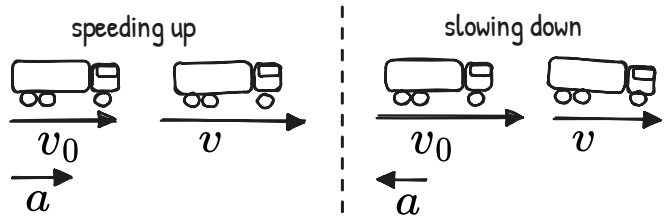
> **Acceleration**
> The **rate at which velocity changes**. Acceleration, a, is a vector measured in $\frac{\text{m}}{\text{s}^2}$.

$$a = \frac{\Delta v}{t} = \frac{v - v_0}{t}$$

An object accelerates when it **speeds up**, **slows down**, or **changes direction**

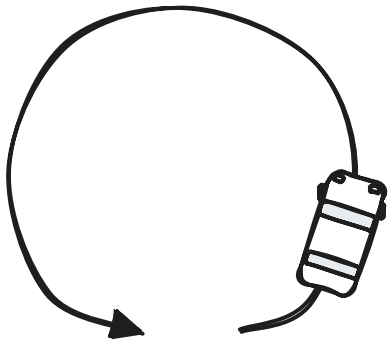
Speeding Up vs Slowing Down

- Velocity and acceleration in same direction: speeding up.
- Velocity and acceleration in opposite direction: slowing down.



A Change in Direction is a Change in Velocity

- Velocity is a **vector**: it has magnitude **and** direction.
- Changing **either one** changes the velocity.
- A car going around a roundabout at a constant speed does *not* have a constant velocity:



Kinematics — Position-Time Graphs

Lesson 4 of 5

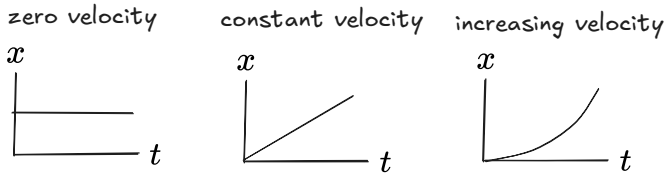
Read position-time graphs to find position, use the slope to find velocity, and describe speeding up, slowing down, and changing direction.

Motion Graphs

The motion of an object can be shown on a graph with **position**, x, on the vertical axis and **time**, t, on the horizontal axis:

- **Position-time graph** — how position changes with time

Position-Time Graphs



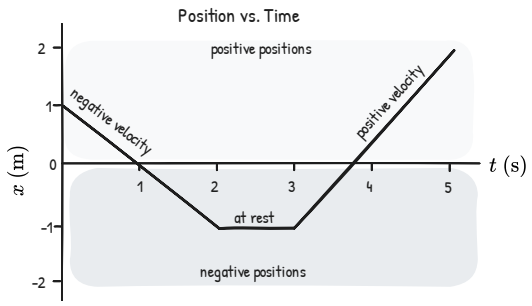
> **Tip**
> On a position-time graph, the value on the vertical axis is the object's **position**, x.

Velocity from a Position-Time Graph

> **Important**
> On a **position-time** graph, the **slope** is the **velocity**: $v = \frac{\Delta x}{t} = \frac{x - x_0}{t}$ (rise over run).

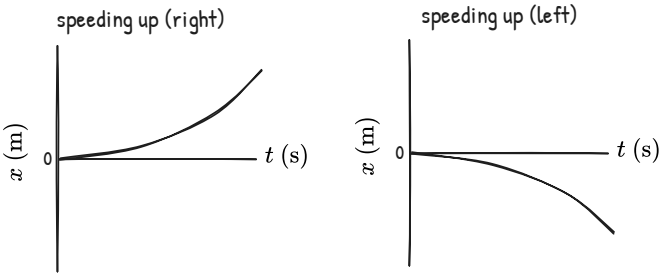
- A steeper slope means a **greater velocity**.
- Positive slope: $v \rightarrow$ or $v \uparrow$
- Negative slope: $v \leftarrow$ or $v \downarrow$

Features of a Position-Time Graph



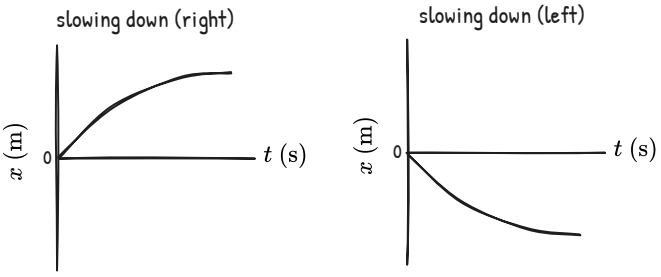
Position-Time: Speeding Up

Speeding up: the slope gets steeper.

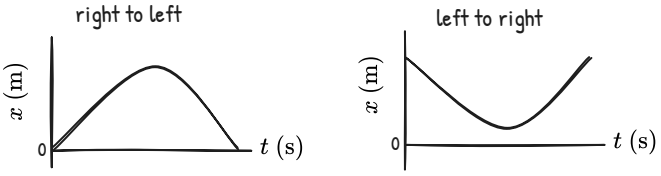


Position-Time: Slowing Down

Slowing down: the slope gets flatter.



Position-Time: Changing Direction



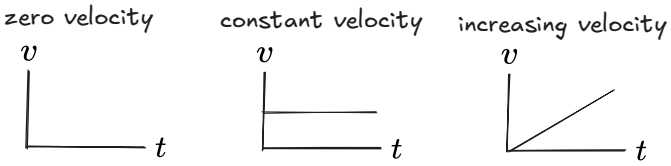
> **Warning**
> If slope changes **positive to negative** (or the reverse), then the object is **changing direction**.

Kinematics — Velocity-Time Graphs

Lesson 5 of 5

Read velocity-time graphs, use the slope to find acceleration, and find displacement from the area under the graph, including its sign and units.

Velocity vs Time



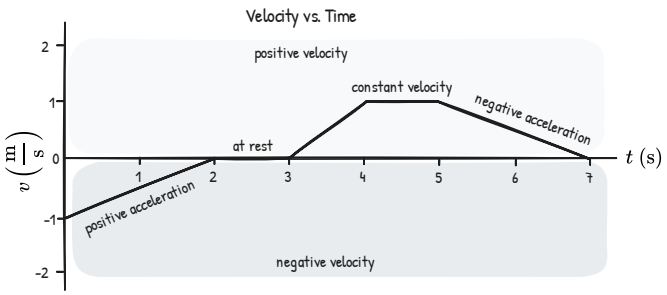
> **Tip**
> On a velocity-time graph, the **velocity**, v, is on the vertical axis.

Acceleration from a Velocity-Time Graph

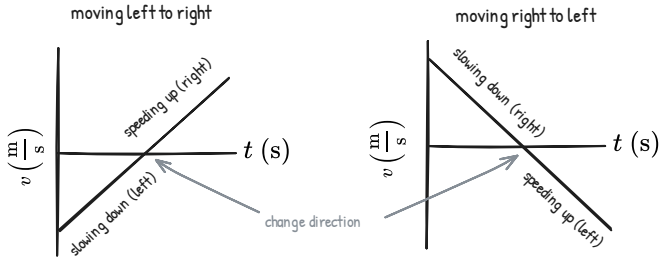
> **Important**
> On a **velocity-time** graph, the **slope** is the **acceleration**: $a = \frac{\Delta v}{t} = \frac{v - v_0}{t}$ (rise over run).

- A steeper slope means a **greater acceleration**.
- Positive slope: $a \rightarrow$ or $a \uparrow$
- Negative slope: $a \leftarrow$ or $a \downarrow$

Features of a Velocity-Time Graph

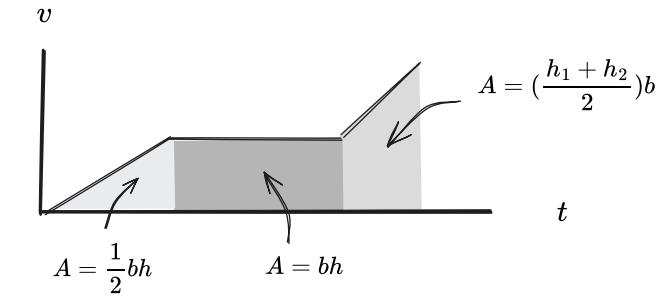


Velocity-Time: Changing Speed and Direction



Area Under the Velocity-Time Graph

> **Important**
> The **area** between the line on a **velocity-time** graph and the horizontal axis is equal to the **displacement**, Δx.



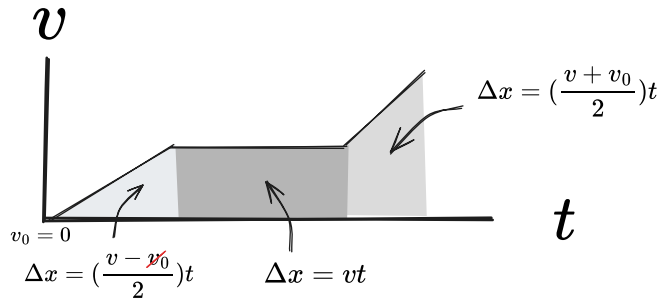
Units of Area Under the Curve

$$\text{area} = \text{velocity} \cdot \text{time} = \frac{\text{m}}{\text{s}} \cdot \text{s} = \text{m}$$

- The unit is **meter** which is the unit of **displacement**.

Area with Kinematic Equations

- Each shape corresponds to a kinematic equation. Notice the similarity between the triangle and trapezoid.



Positive and Negative Displacement

> **Tip**
> Area **above** the axis is **positive** displacement; area **below** the axis is **negative** displacement. Add them to get the total displacement.

